CSA6102 – DIGITAL FORENSICS

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EXP 2

1. Write a Python program to verify the chain of custody of digital evidence using file hashing

```python
# ==========================================================
# Chain of Custody Verification using File Hashing (SHA-256)
# Google Colab Compatible
# ==========================================================

import hashlib

# ----------------------------------------------------------
# Function to calculate SHA-256 hash of a file
# ----------------------------------------------------------
def calculate_hash(filename):
    sha256 = hashlib.sha256()

    with open(filename, "rb") as file:
        while True:
            data = file.read(4096)
            if not data:
                break
            sha256.update(data)

    return sha256.hexdigest()

# ----------------------------------------------------------
# Step 1: Create a digital evidence file
# ----------------------------------------------------------
filename = "digital_evidence.txt"

with open(filename, "w") as file:
    file.write("Digital Evidence\n")
    file.write("Case ID: 101\n")
    file.write("Collected by: Investigator A\n")

print("Evidence file created successfully.\n")

# ----------------------------------------------------------
# Step 2: Generate the original hash
# ----------------------------------------------------------
original_hash = calculate_hash(filename)

print("========== ORIGINAL HASH ==========")
print(original_hash)

# ----------------------------------------------------------
# Step 3: Verify the file (Before Tampering)
# ----------------------------------------------------------
print("\n========== VERIFYING CHAIN OF CUSTODY ==========")

current_hash = calculate_hash(filename)
```

```python
if current_hash == original_hash:
    print("Verification Successful")
    print("Chain of Custody Maintained")
else:
    print("Verification Failed")
    print("Evidence Integrity Compromised")

# ----------------------------------------------------------
# Step 4: Simulate Tampering
# ----------------------------------------------------------
print("\nSimulating Unauthorized Modification...\n")

with open(filename, "a") as file:
    file.write("Tampered Data Added\n")

# ----------------------------------------------------------
# Step 5: Verify Again
# ----------------------------------------------------------
tampered_hash = calculate_hash(filename)

print("========== NEW HASH ==========")
print(tampered_hash)

print("\n========== VERIFYING AFTER TAMPERING ==========")

if tampered_hash == original_hash:
    print("Verification Successful")
    print("Chain of Custody Maintained")
else:
    print("Verification Failed")
    print("Evidence Has Been Modified")
    print("Chain of Custody Broken")

# ----------------------------------------------------------
# Step 6: Display File Contents
# ----------------------------------------------------------
print("\n========== FILE CONTENT ==========")

with open(filename, "r") as file:
    print(file.read())

# ----------------------------------------------------------
# Step 7: Conclusion
# ----------------------------------------------------------
print("========== CONCLUSION ==========")
print("Chain of Custody ensures that digital evidence")
print("remains unchanged from collection to presentation.")
print("SHA-256 hashing helps detect any unauthorized")
print("modification of the evidence.")
```

OUTPUT :

```
••• Evidence file created successfully.

    ========== ORIGINAL HASH ==========
    42d44b6428814a7b37333576ec831adede47f67b06e420db5b323cc3875280dc

    ========== VERIFYING CHAIN OF CUSTODY ==========
    Verification Successful
    Chain of Custody Maintained

    Simulating Unauthorized Modification...

    ========== NEW HASH ==========
    2292136436cb732b6ad2fe87486425b4051e7b097b174be2f99eaf9fd1f1872d

    ========== VERIFYING AFTER TAMPERING ==========
    Verification Failed
    Evidence Has Been Modified
    Chain of Custody Broken

    ========== FILE CONTENT ==========
    Digital Evidence
    Case ID: 101
    Collected by: Investigator A
    Tampered Data Added

    ========== CONCLUSION ==========
    Chain of Custody ensures that digital evidence
    remains unchanged from collection to presentation.
    SHA-256 hashing helps detect any unauthorized
    modification of the evidence.
```